

Unity Orb Experience

Teammate Handoff, Inspector Reference, Setup Guide, and Troubleshooting Manual

Purpose. This document explains how the two orb scripts work together, how every Inspector value changes the experience, how to safely modify the hierarchy, and how to diagnose the most common failures. It is written for a teammate who did not participate in the original development.

Files delivered with this guide:

- **OrbLureController_COMMENTED.cs** - controls the entrance, world-space travel, final stop, proximity check, and handoff.
- **A_OrbIntroSequence_COMMENTED.cs** - controls growth, charged spin, doppelganger reveal, shrinking, chest absorption, disappearance, and optional lowering.
- **Unity_Orb_Experience_Teammate_Guide.pdf** - this reference document.

Critical file rule: The C# filename must match the class name. Rename the delivered files to **OrbLureController.cs** and **A_OrbIntroSequence.cs** before placing them in Assets. Keep only one copy of each class in the project. Files named UPDATED, v2, v3, backup, or copy can still compile the same class and cause duplicate-symbol Android build failures.

1. What the Experience Does

The experience is divided into two scripts so the entrance can be tuned separately from the manifestation sequence.

Phase A - OrbLureController

- Waits for the opening delay.
- Locks the participant's starting head position and horizontal facing direction.
- Places the orb behind the participant.
- Moves it into the upper-right area, pauses, travels forward on the right, then travels toward center.
- Stops in world space and waits for the participant to approach.
- Starts A_OrbIntroSequence when the participant is close enough or the timeout expires.

Phase B - A_OrbIntroSequence

- Ramps spin speed and grows the orb.
- Moves the orb backward horizontally as its radius grows so its surface does not crowd the participant.
- Keeps the bottom of the orb above the floor within a capped vertical correction.
- Fades the doppelganger into the center of the orb while keeping it facing the participant.
- Shrinks the orb in front of the chest.
- Keeps the small orb spinning while it enters the chest, then hides it immediately.
- Optionally lowers the doppelganger after absorption without rotating it.

2. Required Hierarchy

```
INTRO_VR_WORLD
  Neb_orb_Pivot                <- OrbLureController reference: Orb Pivot
    Neb_orb_VisualPivot        <- visual only; spins, scales, then hides
      Orb mesh / particles / video
    OrbSequenceController      <- A_OrbIntroSequence component can live here
    Doppelganger               <- must NOT be inside Neb_orb_VisualPivot
      Armature
        Chest / UpperChest / Spine2 <- Doppelganger Chest reference

[BuildingBlock] Camera Rig
  TrackingSpace
    CenterEyeAnchor            <- Player Head reference for both scripts
```

Why the doppelganger must be outside Neb_orb_VisualPivot: the visual pivot spins and scales. If the doppelganger is its child, the model will spin, inherit the orb scale, or disappear when the visual pivot is disabled.

3. Component Placement and Reference Assignments

OrbLureController normally goes on Neb_orb_Pivot.

Inspector field	What it changes	How to adjust it
Orb Pivot	The world-space movement pivot for the entire orb sequence.	Assign Neb_orb_Pivot.
Spin Visual	The child that rotates during lure movement.	Assign Neb_orb_VisualPivot. Do not assign the doppelganger.
Player Head	The participant camera position and direction.	Assign CenterEyeAnchor.
Orb Sequence	The second script that begins after approach.	Assign the A_OrbIntroSequence component.

Inspector field	What it changes	How to adjust it
Orb Root To Show Hide	The visual object hidden before the entrance and shown when movement begins.	Assign Neb_orb_VisualPivot.

A_OrbIntroSequence normally goes on OrbSequenceController or another stable object under Neb_orb_Pivot.

Inspector field	What it changes	How to adjust it
Neb Orb	The moving world-space pivot.	Assign Neb_orb_Pivot.
Spin Visual	The child that spins and scales.	Assign Neb_orb_VisualPivot.
Player Head	Used for distance, facing, and relative placement.	Assign CenterEyeAnchor.
Doppelganger Root	The complete model root moved during centering and lowering.	Assign Doppelganger.
Doppelganger Chest	The bone used for alignment and absorption.	Assign Chest, UpperChest, or Spine2 inside the model.
Turn Off On Absorb	The object hidden when the orb reaches the chest.	Assign Neb_orb_VisualPivot only.

4. OrbLureController - Every Inspector Field

Inspector field	What it changes	How to adjust it
Start Delay Seconds	Delay before the orb is placed behind the participant and shown.	Increase for a longer quiet opening; set to 0 for immediate entrance.
Behind Meters	How far behind the starting head position the orb begins.	Increase to start farther behind; decrease for a shorter reveal path.
Right Meters	Sideways distance of the upper-right and right-travel path.	Increase to push farther right; decrease to keep closer to center.
Right Forward Meters	Forward distance of the first visible upper-right point.	Increase to make the reveal farther away; decrease to bring it closer.
Height Above Head	Vertical offset for the behind, upper-right, and right-travel points.	Increase to place the entrance higher; decrease to place it lower.
Entrance Duration Seconds	Travel time from behind to upper-right.	Increase for slower entrance; decrease for a faster reveal.
Right Hold Seconds	Pause length at the upper-right point.	Increase for more time to notice it; set to 0 to remove the pause.
Right Travel Forward Meters	Forward endpoint of the long right-side movement.	Increase for a longer path into the room; decrease for a shorter path.
Right Travel Duration Seconds	Time spent moving along the right side.	Increase for slower guiding motion; decrease for faster movement.
Final Forward Meters	Forward distance of the final center stop from the starting head.	Increase to make the participant walk farther; decrease for a closer stop.
Final Right Meters	Side offset of the final stop.	0 centers it; positive moves right; negative moves left.
Final Height Above Start Head	Final stop height relative to the starting head position.	Increase to stop higher; decrease to stop lower. This is not floor-relative.
Center Travel Duration Seconds	Time moving from the right-side endpoint to the final center point.	Increase for a slower center transition.
Idle Spin Degrees Per Second	Spin speed during lure movement and final waiting.	Increase for faster idle rotation; decrease for calmer motion.
Spin Axis Local	Local axis used to rotate the visual.	Usually (0,1,0). Change only when the visual model uses a different orientation.
Hover Amplitude	Maximum vertical hover distance in meters.	Increase for stronger bobbing; set to 0 for no hover.
Hover Speed	Frequency of the vertical hover wave.	Increase for quicker bobbing; decrease for slower floating.
Minimum Spacing Meters	Minimum center-to-head distance enforced during lure movement.	Increase to keep the orb farther away. This does not measure the orb surface.
Spacing Correction Speed	How strongly the script reacts when the participant is too close.	Increase for faster correction; lower values feel softer.
Maximum Correction Meters Per Second	Caps the spacing push so it cannot jump too quickly.	Increase only if spacing correction is too slow; decrease if motion looks abrupt.
Required Distance To Start	Head-to-orb distance that triggers the manifestation sequence.	Increase to trigger sooner/farther away; decrease to require a closer approach.
Force Start After Seconds	Safety timeout after the orb reaches its final point.	Increase for more participant-controlled timing; lower for guaranteed pacing.
Auto Start	Whether the lure begins automatically in Start().	Keep checked for normal scene launch. Uncheck only when another controller calls BeginLure().

5. A_OrbIntroSequence - Every Inspector Field

World Height and Floor Protection

Inspector field	What it changes	How to adjust it
Minimum Orb Center Y	Absolute lowest world Y allowed for the orb pivot center.	Raise when the scene floor is above world Y=0 or clipping remains.
Minimum Doppel Root Y	Absolute lowest world Y allowed for the doppelganger root.	Raise only if the model root can fall below the scene floor.
Keep Orb Above Ground	Turns radius-aware floor protection on during growth.	Keep checked for normal use.
Orb Ground Clearance	Extra air gap between the calculated orb bottom and the floor.	Increase when the orb slightly intersects the ground.
Maximum Ground Lift During Growth	Caps how far floor protection can raise the orb center.	Increase if clipping remains; decrease if the orb rises too much.

Doppelganger Fade and Facing

Inspector field	What it changes	How to adjust it
Doppel Fade Duration	Time required to fade from invisible to fully visible.	Increase for slower materialization; decrease for a faster reveal.
Doppel Fade Curve	Shape of alpha change over time.	Edit curve handles to delay the reveal, accelerate it, or soften the ending.
Keep Doppel Centered In Orb	Continuously aligns the chest to the orb center during the fade.	Keep checked unless the model should stay at a manually placed location.
Doppel Local Offset	Offset from orb center used when aligning the chest.	More negative Y lowers the model inside the orb; X/Z shift sideways/depth.
Face Player During Fade	Keeps the model facing the participant while appearing.	Keep checked for direct confrontation; uncheck for fixed orientation.
Doppel Facing Yaw Offset	Corrects imported model forward direction.	Try 0, 90, -90, or 180 until the front of the model faces the participant.

Full-Size Holds and Events

Inspector field	What it changes	How to adjust it
Full Size Hold Before Fade	Pause after growth, before the doppelganger begins appearing.	Increase for a dramatic charged-orb hold.
Full Size Hold After Fade	Pause after full visibility, before shrinking begins.	Increase to let the participant observe the model longer.
On Doppelganger Fade In	UnityEvent invoked immediately before fade animation.	Use for audio, particles, lighting, or timeline triggers.
On Absorb Start	UnityEvent invoked before the shrink/absorption sequence.	Use for sound, chest effects, or scene-state changes.

Optional Standalone Start

Inspector field	What it changes	How to adjust it
Start Forward	Forward offset used only when this script starts itself.	Ignored when OrbLureController hands off from the current position.
Start Height	Vertical offset used only for standalone auto-start.	Ignored in normal two-script operation.
Start Right	Side offset used only for standalone auto-start.	Ignored in normal two-script operation.
Snap Orb On Start	Moves the orb to the standalone start point before beginning.	Use only for isolated testing of the second script.

Growth

Inspector field	What it changes	How to adjust it
Grow And Spin Up Duration	Total growth and spin-ramp time.	Increase for slower transformation; decrease for faster charging.
Grow Hover Amplitude	Small vertical hover applied during growth.	Keep low to avoid noticeable floor movement; set to 0 for no hover.
Grow Hover Speed	Speed of growth-phase hover.	Increase for quicker oscillation.
Grow Extra Surface Gap	Extra protected space added between the participant and orb surface.	Increase if the full-size orb still feels too close.
Grow Back Away Smoothing	Responsiveness of backward movement while radius increases.	Increase to follow the calculated safe distance faster; decrease for softer lag.

Heartbeat Pulse

Inspector field	What it changes	How to adjust it
Heartbeat Strength	Amount of temporary scale added by the pulse.	Increase carefully; large values can make the orb visibly jump.
Heartbeat Rate	Number of heartbeat-curve cycles per second.	Increase for a faster pulse rhythm.
Heartbeat Curve	Shape of one repeating pulse cycle.	Edit keyframes for single beat, double beat, or long pauses.

Spin and Scale

Inspector field	What it changes	How to adjust it
Spin Axis Local	Local axis used to rotate the visual.	Usually (0,1,0).
Spin Idle	Starting spin speed at the beginning of growth.	Match or approximate the lure script idle speed.
Spin Charged	Final charged spin speed.	Increase for stronger energy; reduce if visually uncomfortable.
Spin Ramp	How spin transitions from idle to charged.	Use a slow early curve and steep ending for dramatic acceleration.
Scale Start	Visual scale at the start of growth.	Usually 1 when the model is authored at normal size.
Scale Charged	Final full orb scale.	Increase for a larger manifestation volume; affects floor and spacing calculations.
Scale Ramp	How scale changes over the growth duration.	Edit curve for delayed growth, fast expansion, or gentle easing.

Orb Size and Participant Spacing

Inspector field	What it changes	How to adjust it
Base Orb Radius Meters	Approximate visible radius when visual scale equals 1.	Measure/tune until floor and surface-distance calculations match the actual mesh.
Camera Safety Margin Meters	Minimum protected gap between head and nearest orb surface.	Increase for comfort; decrease only after headset testing.

Shrink in Front of Chest

Inspector field	What it changes	How to adjust it
Shrink Duration	Time to move toward the chest-front point and reach Small Orb Scale.	Increase for slower compression; decrease for a quicker collapse.
Small Orb Scale	Scale reached before entering the chest.	Lower for a tinier final orb; avoid exactly 0 before absorption.
Shrink Front Of Chest Meters	Distance from chest toward the participant where shrinking ends.	Increase to finish farther in front; decrease to finish closer to the chest.
Shrink Front Of Chest Height	Vertical offset from the chest-front point.	Positive moves upward; negative moves downward.
Shrink Curve	Movement/scale easing for the shrink phase.	Adjust curve to change where most of the shrinking happens.

Chest Absorption

Inspector field	What it changes	How to adjust it
Absorb Duration	Time for the small orb to move into the chest and disappear.	Increase for slower entry; decrease for faster absorption.
Absorb Target	Optional fallback Transform when no chest bone is assigned.	Normally leave empty when Doppelganger Chest is correctly assigned.
Absorb Forward	Participant-relative fallback forward distance.	Used only when both chest and absorb target are missing.
Absorb Height	Participant-relative fallback vertical offset.	Used only in the final fallback case.

Post-Absorb Doppel Lowering

Inspector field	What it changes	How to adjust it
Lower Doppel After Absorb	Enables vertical adjustment after the orb has disappeared.	Uncheck to leave the model exactly where it manifested.
Doppel Chest Y Offset From Head	Desired chest height relative to participant head Y.	More negative lowers the model farther; less negative keeps it higher.
Post Absorb Lower Duration	Time for the vertical lowering/raising motion.	Increase for a slower settling movement.

Debug

Inspector field	What it changes	How to adjust it
Auto Start	Starts A_OrbIntroSequence directly in Start().	Keep unchecked when OrbLureController is used. Check only for isolated testing.

6. Common Changes and Exactly What to Adjust

Desired change	Adjustment
Orb should appear farther right	Increase Right Meters in OrbLureController.
Orb should appear higher	Increase Height Above Head. For final stop only, increase Final Height Above Start Head.
Orb should guide the participant farther	Increase Final Forward Meters and possibly Right Travel Forward Meters.
Entrance is too fast	Increase Entrance Duration, Right Travel Duration, and/or Center Travel Duration.
Orb is too close while waiting	Increase Minimum Spacing Meters.
Growth orb surface feels too close	Increase Grow Extra Surface Gap and/or Camera Safety Margin Meters.
Orb intersects the floor	Increase Base Orb Radius Meters, Orb Ground Clearance, Minimum Orb Center Y, or Maximum Ground Lift During Growth.
Orb rises too much during growth	Decrease Maximum Ground Lift During Growth or verify Base Orb Radius is not too large.
Doppelganger faces sideways	Change Doppel Facing Yaw Offset among 0, 90, -90, and 180.
Doppelganger is too high inside orb	Make Doppel Local Offset Y more negative.
Doppelganger lowers too much after absorption	Make Doppel Chest Y Offset From Head less negative or disable Lower Doppel After Absorb.
Reveal is too quick	Increase Doppel Fade Duration and soften the Fade Curve.
Orb should shrink more before entering chest	Lower Small Orb Scale.
Small orb should enter faster	Decrease Absorb Duration.
Orb does not disappear	Assign Turn Off On Absorb to Neb_orb_VisualPivot.
Doppelganger disappears with orb	Turn Off On Absorb is assigned too high in the hierarchy. Assign only Neb_orb_VisualPivot.

7. Safe Testing Workflow

- Duplicate the scene before major changes.
- Verify there is only one active camera rig and one hand/input rig.
- Keep A_OrbIntroSequence Auto Start unchecked and OrbLureController Auto Start checked for the normal flow.
- Clear the Console before each test.
- Test the lure path first. Then walk toward the final orb and observe the full manifestation.
- After changing hierarchy references, exit Play mode and save the scene before building.
- For Quest walking tests, build an APK and install it standalone; this avoids Quest Link Editor instability.

Fast timing test: temporarily shorten travel, growth, fade, shrink, and absorb durations. Restore performance timing before final delivery.

8. Troubleshooting

Symptom	Likely cause / fix
Nothing starts	Confirm OrbLureController component is enabled, Auto Start is checked, Player Head is assigned, and the orb sequence reference is valid.
Orb appears but second sequence never starts	Increase Required Distance To Start or reduce Force Start After Seconds for testing.
Doppelganger position becomes NaN	Check that Neb Orb, Player Head, Doppelganger Root, and Chest transforms have valid finite values. Ensure no parent has zero/NaN scale. Keep the doppelganger outside the spinning/scaling visual pivot.
Doppelganger spins	It is likely parented under Neb_orb_VisualPivot, or Spin Visual references the wrong Transform.
Doppelganger vanishes after absorption	Turn Off On Absorb references a parent that contains the doppelganger.
Pink/magenta materials	The shader is missing or incompatible with the active render pipeline. Inspect the material shader and URP pipeline asset.
Android duplicate symbol A_OrbIntroSequence	More than one C# file declares public class A_OrbIntroSequence. Search all Assets and keep only one.
Android duplicate symbol OrbLureController	More than one C# file declares public class OrbLureController. Keep only one.
Unity takes a long time after deleting Library/Bee	Unity is rebuilding generated IL2CPP/Bee caches. Let it finish; project assets were not deleted.
Build uses old scene	Check Build Profiles > Scene List. Save the current scene and project before building.

9. Build and Handoff Checklist

- Rename delivered scripts to OrbLureController.cs and A_OrbIntroSequence.cs.
- Delete or move all older copies outside the Assets folder.
- Confirm no Console compilation errors.
- Confirm hierarchy and all references listed in Sections 2 and 3.
- Confirm Neb_orb_VisualPivot is the only object assigned to Turn Off On Absorb.
- Confirm the doppelganger is not a child of Neb_orb_VisualPivot.
- Confirm OrbLureController Auto Start is checked.
- Confirm A_OrbIntroSequence Auto Start is unchecked.
- Save the scene and Save Project.
- Confirm the correct scene is first and enabled in the Android build profile.
- Build an APK for ARM64/IL2CPP and install it to the Quest for a standalone walking test.

Recommended source-control handoff: include Assets, Packages, and ProjectSettings. The Library folder is generated and can be omitted, but opening without it will trigger a long reimport.

10. Script Interaction Summary

```
OrbLureController.Start()  
-> hides orb visual  
-> RunLure()  
    -> lock starting pose  
    -> build fixed world-space path  
    -> entrance / hold / travel / center  
    -> wait for approach or timeout  
    -> orbSequence.BeginMainSequenceFromCurrentPosition()  
  
A_OrbIntroSequence.MainSequence()  
-> grow + spin up + back away + floor protection  
-> optional hold  
-> fade doppelganger in and face participant  
-> optional hold  
-> shrink orb in front of chest  
-> spin orb into chest  
-> hide Neb_orb_VisualPivot immediately  
-> optionally lower doppelganger vertically
```

The code comments in the delivered scripts explain each method, its responsibility, and the hierarchy assumptions that prevent spinning, scaling, disappearance, and positioning problems.